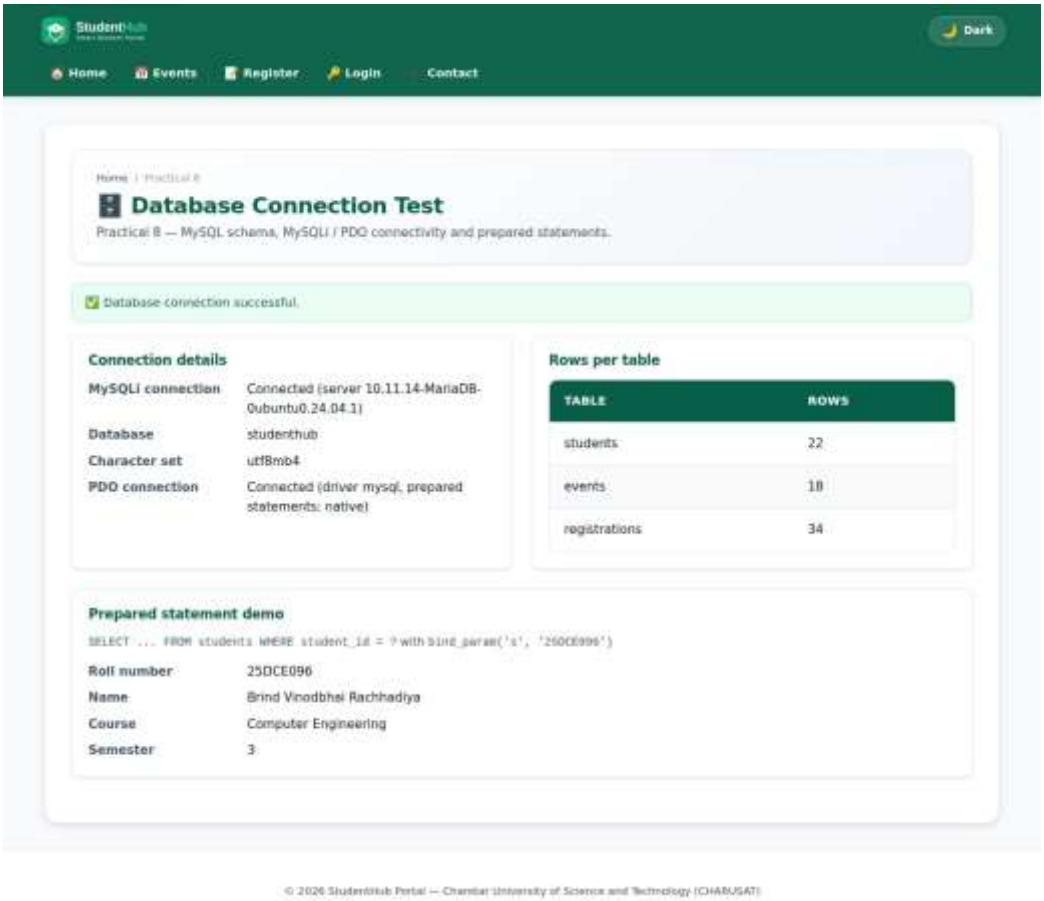
PRACTICAL-8

AIM: Design the StudentHub database (students, events and registrations) with an ER model, create the MySQL schema with primary and foreign keys, and connect PHP to MySQL using MySQLi and PDO with prepared statements.

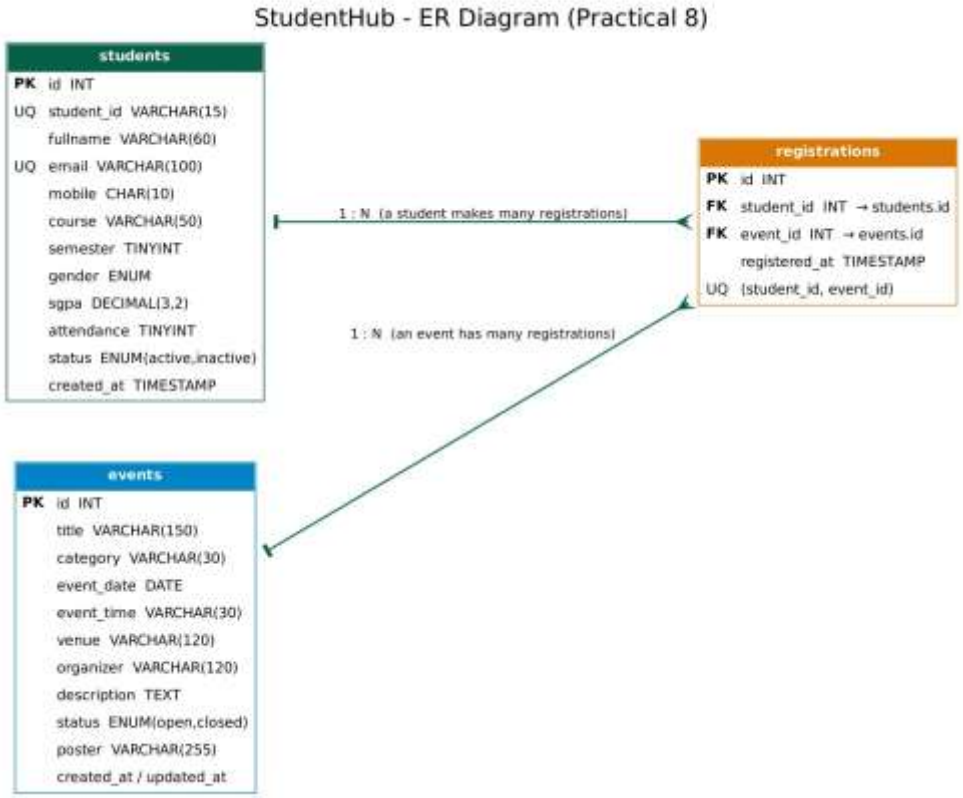
SOURCE CODE :

https://github.com/rachhadiyabrind-del/WDF_B/tree/main/Practical(1-4)(Studenthub)/Practical-8

PREVIEW:



Connection test page - MySQLi and PDO both connected



ER diagram - students, events and registrations

```
mysql> SHOW TABLE STATUS (database: studenthub)
```

Table	Engine	Rows	Collation
audit_logs	InnoDB	0	utf8mb4_unicode_ci
events	InnoDB	2	utf8mb4_unicode_ci
registrations	InnoDB	2	utf8mb4_unicode_ci
remember_tokens	InnoDB	0	utf8mb4_unicode_ci
students	InnoDB	22	utf8mb4_unicode_ci
users	InnoDB	1	utf8mb4_unicode_ci

Tables created from schema.sql

```
mysql> foreign keys of studenthub
```

Constraint_Name	Table	Column_Name	Ref_Table	Ref_Column
fk_audit_user	audit_logs	user_id	users	id
fk_reg_event	registrations	event_id	events	id
fk_reg_student	registrations	student_id	students	id
fk_remember_user	remember_tokens	user_id	users	id

Foreign keys on the registrations table

TEST CASES :

#	Input / action	Expected result	Result
1	Open test_connection.php	"Database connection successful"; MySQLi and PDO both shown as connected; row counts listed	Pass
2	Import schema.sql and seed.sql	Tables students, events and registrations created with	Pass

		seed rows	
3	Inspect the ER diagram	Three entities; registrations resolves the many-to-many link between students and events	Pass
4	Inspect foreign keys on registrations	fk_reg_student and fk_reg_event point to students.id and events.id	Pass
5	Insert the same student and event pair twice	Second insert rejected by the unique key uq_registration	Pass
6	Insert a registration with a non-existent student_id	Rejected by the foreign key constraint	Pass
7	Call sp_register_student_for_event for a closed event	result_code = 1; no row inserted	Pass
8	Use a wrong database password	Generic "database temporarily unavailable" message; real error written only to the log	Pass

KEY QUESTIONS :

1. Is the schema normalized properly?

Yes, it is in third normal form. Each table stores one kind of fact: student details live only in students and event details only in events. The many-to-many relation between students and events is split into the registrations table, so there are no comma-separated lists or repeated groups of columns, and every non-key column depends only on its own table's primary key.

2. Are primary keys and foreign keys correctly defined?

Each table has an AUTO_INCREMENT primary key (students.id, events.id, registrations.id). registrations.student_id and registrations.event_id are foreign keys to students.id and events.id, with ON DELETE CASCADE, so deleting a student or event removes its registrations. A composite UNIQUE key on (student_id, event_id) stops a student from registering for the same event twice. students.student_id and students.email are also unique.

```
UNIQUE KEY uq_registration (student_id, event_id)
```

3. Is the database connection handled securely?

Yes. mysqli_report() is set to strict mode, so every database error becomes an exception. Credentials come from environment variables with XAMPP defaults rather than being typed into every page. Error details go only to the server log; the visitor sees a generic 503 message. The connection uses utf8mb4 to match the tables. The PDO version turns off emulated prepares (PDO::ATTR_EMULATE_PREPARES = false), so queries use real server-side prepared statements, and the test page binds the roll number as a parameter instead of concatenating it into the SQL.